



Contents lists available at ScienceDirect

The Journal of Arthroplasty

journal homepage: www.arthroplastyjournal.org

Primary Knee

Coronal and Sagittal Component Position in Contemporary Total Knee Arthroplasty: Targeting Native Alignment Optimizes Clinical Outcomes



Hassan Farooq, MD ^a, Evan R. Deckard, BSE ^b, Justin Carlson, BS ^c,
Nathan Ghattas, BS ^c, R. Michael Meneghini, MD ^{b,d,*}

^a Department of Orthopaedic Surgery and Rehabilitation, Loyola University Medical Center, Maywood, Illinois^b Indiana Joint Replacement Institute, Indianapolis, Indiana^c Department of Mechanical Engineering, Tennessee Technological University, Cookeville, Tennessee^d Department of Orthopaedic Surgery, Indiana University School of Medicine, Indianapolis, Indiana

ARTICLE INFO

Article history:

Received 1 December 2022

Received in revised form

6 April 2023

Accepted 16 April 2023

Available online 24 April 2023

Keywords:

total knee arthroplasty

machine learning

alignment

three-dimensional

patient reported outcomes

minimal clinically important difference

<https://www.aahks.org>

ABSTRACT

Background: Advanced technologies, like robotics, provide enhanced precision for implanting total knee arthroplasty components; however, optimal component position and limb alignment remain unknown. This study sought to identify sagittal and coronal alignment targets that correlate with minimal clinically important differences (MCIDs) in patient-reported outcome measures (PROMs).

Methods: A total of 1,311 consecutive total knee arthroplasties were retrospectively reviewed. Posterior tibial slope (PTS), femoral flexion (FF), and tibio-femoral alignment (TFA) were measured radiographically. Patients were grouped based on whether they achieved multiple MCIDs for PROM scores. Classification and regression tree machine learning models were utilized to identify optimal alignment zones. The mean follow-up was 2.4 years (range, 1 to 11).

Results: The change in PTS and postoperative TFA were most predictive for achieving MCIDs in 90% of the models. Approximating native PTS within 4° correlated with MCID achievement and superior PROMs. Preoperative varus and neutral aligned knees were more likely to meet MCIDs and superior PROM scores when not overcorrected into valgus postoperatively ($\geq 7^\circ$). Preoperative valgus-aligned knees correlated with MCID achievement when postoperative TFA was not overcorrected into substantial varus ($< 0^\circ$). Albeit less impactful, FF $\leq 7^\circ$ correlated with MCID achievement and superior PROMs regardless of preoperative alignment. Sagittal and coronal alignment measurements had moderate to strong interactions in 13 of 20 models.

Conclusion: Optimized PROM MCIDs correlated with approximating native PTS while maintaining similar preoperative TFA and incorporating moderate FF. Study findings demonstrate interactions between sagittal and coronal alignment which may optimize PROMs, highlighting the importance of three-dimensional implant alignment targets.

Level of Evidence: III.

© 2023 Elsevier Inc. All rights reserved.

Satisfaction following primary total knee arthroplasty (TKA) remains stagnant at approximately 80%, leaving 1 in 5 patients unsatisfied with their procedure [1–3]. Several nonmodifiable factors

are reported to predict satisfaction after TKA [3–5]. However, other modifiable factors such as implant alignment have been reported to predict satisfaction and other patient-reported outcome measures (PROMs) [6–9]. Recently, systematic algorithms related to variations of mechanical [10] and kinematic alignment [10,11], and now gait phenotyping [12,13] have increased in usage to address the plateaued state of satisfaction and functional outcomes following TKA. While mechanical and kinematic alignment techniques report great to excellent outcomes, recent studies report no significant difference between the two techniques [11,14–16], with no improvement beyond the 80% satisfaction rate. More recently, the

One or more of the authors of this paper have disclosed potential or pertinent conflicts of interest, which may include receipt of payment, either direct or indirect, institutional support, or association with an entity in the biomedical field which may be perceived to have potential conflict of interest with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.arth.2023.04.040>.

* Address correspondence to: R. Michael Meneghini, MD, Indiana Joint Replacement Institute, 1725 North 5th Street, Terre Haute, IN 47804.

<https://doi.org/10.1016/j.arth.2023.04.040>

0883-5403/© 2023 Elsevier Inc. All rights reserved.

concept of a more personalized approach to TKA alignment based on knee phenotypes has been identified [12,13]; however, studies have yet to evaluate functional outcomes related to these gait phenotypes. In addition, while several studies investigated alignment differences comparing traditional techniques, the majority of studies only evaluated the alignment in the coronal plane and neglected the sagittal plane, which has been reported to correlate with increased satisfaction and a knee 'always feeling normal' [9].

Furthermore, advanced technologies, like robotic assistance, are increasing in popularity due to the reduction of alignment outliers [17,18], and therefore provide surgeons with enhanced precision for implanting TKA components. While surgical precision has shown improvement, data on the true accuracy of optimal implant alignment are lacking, which are necessary for these advanced technologies to become truly valuable. Therefore, this study sought to identify sagittal and coronal alignment target zones that correlate with minimal clinically important differences (MCIDs) in PROMs.

Methods

Upon institutional review board approval, 1,429 primary TKAs performed by one surgeon from September 27, 2010 to July 1, 2019, at a single institution were retrospectively reviewed. A total of 118 cases were excluded due to reasons potentially affecting outcomes as follows: missing or suboptimal radiograph quality ($n = 13$); hybrid cement fixation ($n = 15$); patella left unresurfaced ($n = 15$); medical complication such as dementia, hematologic cancer, Wilson's disease, ileus, etc. ($n = 10$); orthopaedically complex cases which required augments, removal of previous hardware, prior osteotomies, etc. ($n = 17$); cases requiring an additional reoperation or early revision ($n = 37$); and patients lacking PROMs ($n = 11$). After exclusions, 1,311 TKAs were available for analyses.

Surgical Technique

A medial parapatellar approach was used for all cases, and computer navigation was used on most cases for the distal femoral cut. Traditional extramedullary guides were used for the proximal tibial cut. In general, neutral mechanical coronal alignment and femoral flexion of 5 to 6 degrees were targeted for the femoral component position. Posterior-referencing femoral sizing was used in both cases, enacting an accurate measured resection of the posterior condyles thus restoring native posterior femoral condylar offset. Neutral mechanical tibial alignment was targeted in native valgus knees, and moderate constitutional varus up to 2 degrees mechanical varus was targeted in native varus knees using a traditional extramedullary alignment guide. The patients' native tibial slope was targeted based on the visualized topography of the tibial surface both visualized from the front of the knee, as well as visualized from the medial aspect of the knee with a stylus inserted through the cutting guide. One of two knee arthroplasty systems was used for all cases (EMPOWR 3D Knee, Enovis, Wilmington, DL; or Triathlon Total Knee System, Stryker, Mahwah, NJ).

Manual chart review in the electronic medical record was performed to collect study variables of age, patient sex, height, weight, body mass index, implant type, and American Society of Anesthesiologists-physical status classification; and covariates related to PROMs such as lumbar spine disease, fibromyalgia, systemic lupus erythematosus, rheumatoid arthritis, psoriatic arthritis, presence of depression with or without medication control, and preoperative narcotic usage. The cohort was 67% women and 38% American Society of Anesthesiologists-physical status classification I or II with patients having a mean age and body mass index of 66 years (range, 33 to 91) and 34.2 kg/m² (range, 16 to 59), respectively. All TKAs were anterior-lipped (41.5%) or conforming

polyethylene (58.5%) bearing designs. The mean follow-up of the cohort was 2.4 years (range, 1 to 11).

Radiographic measurements of preoperative and postoperative posterior tibial slope (PTS); preoperative and postoperative tibiofemoral alignment (TFA); and postoperative femoral flexion (FF) were measured on short-knee radiographs according to the Knee Society radiographic evaluation system (See Fig. 1) [19]. One experienced rater performed all radiographic measurements after extensive training from research staff and the senior surgeon. A subset of 702 cases was used for measurement reliability by a second independent rater. Rater agreement was 100% within 4 degrees and $\geq 73\%$ within 2 degrees for all tibial slope measurements. The radiographic rater was blinded to all PROMs data. Overall, mean preoperative PTS was 8.7 degrees (range, 0 to 22), mean postoperative PTS was 7.1 degrees (range, -1 to 21), mean change in PTS was -1.6 degrees (range, -14 to 9), mean preoperative TFA was -0.2 valgus (range, -21 to 22), mean postoperative TFA was -3.5 valgus (range, -11 to 11), and mean postoperative FF was 3.3 degrees (range, -10 to 17).

Patients were grouped for analyses by preoperative TFA to account for patient-specific differences in anatomy and alignment. Preoperative valgus knees were defined as < -7 degrees of TFA (16%, $N = 207$), preoperative neutral knees were defined as -7 to -5 degrees (9%, $N = 116$), and preoperative varus knees were defined as ≥ -4 degrees (75%, $N = 988$), as previously described [7].

Primary outcomes were PROMs related to activity level, knee-specific pain and function, global satisfaction, and their respective MCIDs evaluated using a patient's most recent PROMs at a minimum of 1-year follow-up. Preoperative and postoperative activity levels were evaluated with the University of California Los Angeles Activity Level score (MCID 0.92) [20]. Components of the modern Knee Society Score (KSS) related to pain with level walking [21], pain with climbing stairs (MCIDs 2.0) [21], and how normal a knee feels preoperatively and postoperatively (KSS knee normal score) were evaluated. The Knee Injury and Osteoarthritis Outcome Score for Joint Replacement was used to evaluate overall knee health and function (MCID 15.0) [22]. Also, a Likert 5-point scale was used to evaluate patient satisfaction (no reported MCID). The data response rate for PROMs was 83% at a minimum of 1-year follow-up. In addition, there was minimal effect of covariates on PROMs and achievement of PROM MCIDs ($P \geq .168$). For analyses, patients were grouped based on whether they achieved the reported MCIDs and optimal responses for KSS knee normal score and global satisfaction. Figure 2 shows the breakdown of patients achieving different combinations of PROM MCIDs for analyses.

Data Analyses

All data analyses were performed in Minitab 21.2 with the predictive analytics module (State College, Pennsylvania). The Classification and Regression Tree (CART) models [23,24] (example shown in Fig. 3) were utilized to identify optimal alignment zones for achieving individual and multiple PROM MCIDs simultaneously. Validation of CART model results employed a 10-cross-fold validation with Gini node splitting methodology. Outliers were evaluated with Dixon's r22 ratio tests. The means were reported for normally distributed data with confidence intervals (CIs) reported at the 95th percentile, while medians were reported for data with a non-normal distribution with interquartile ranges. Grouping patients by achieving multiple MCIDs simultaneously provided a more even split of positive and negative binary outcome values and therefore did not require any data up-sampling. In addition, CART models assigned each model variable a relative importance (RI) score based on its predictive ability on the given outcome [met MCID(s) versus did not]. The CART models, compared to traditional

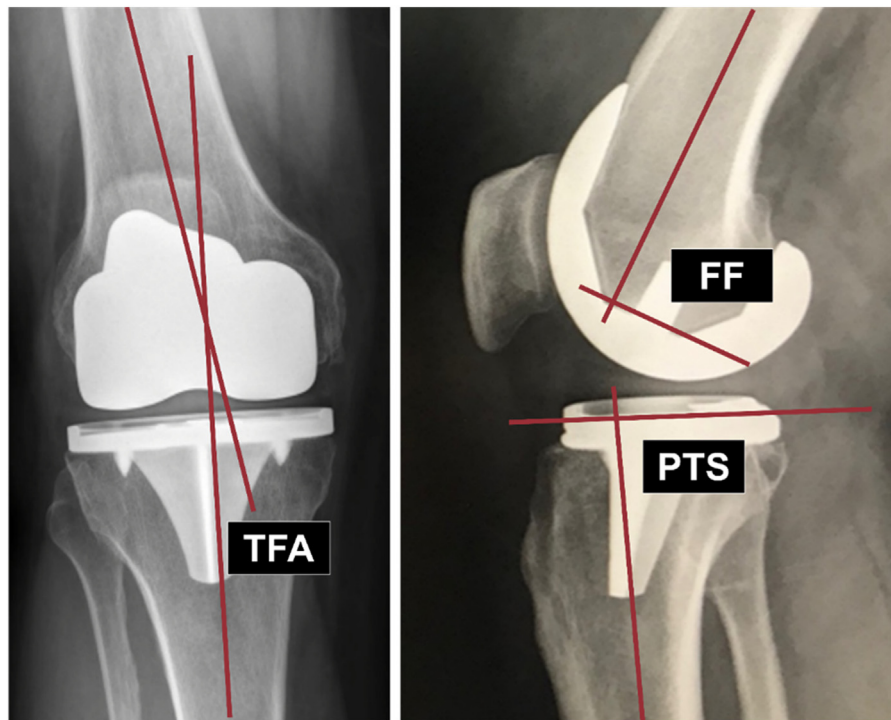


Fig. 1. Measurement techniques for preoperative and postoperative tibio-femoral alignment (TFA), preoperative and postoperative posterior tibial slope (PTS), and postoperative femoral flexion (FF) on short-knee leg radiographs.

statistical methods, allow for the identification of complex interactions typically overlooked in traditional statistical models. Furthermore, CART models allow for a more “data-driven” approach to the model compared to traditional statistical modeling, which “fit data” to a specific pattern (linear regression, parabolic behavior, etc.). However, machine learning algorithms are prone to data overfitting [4,25].

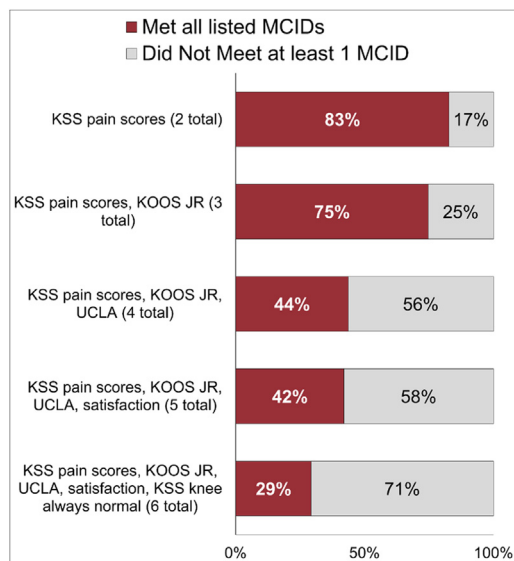


Fig. 2. Data division showing the combinations of achieving multiple minimal clinically important differences (MCIDs). For example, “Knee Society Score (KSS) pain scores” represent patients that met both the MCID for pain with level walking and pain with climbing stairs. Furthermore, “KSS pain scores, Knee Injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS JR), University of California Los Angeles (UCLA) activity level, satisfaction, KSS knee always normal” represents patients that met every MCID and reported all satisfactory responses as the “best case scenario.”

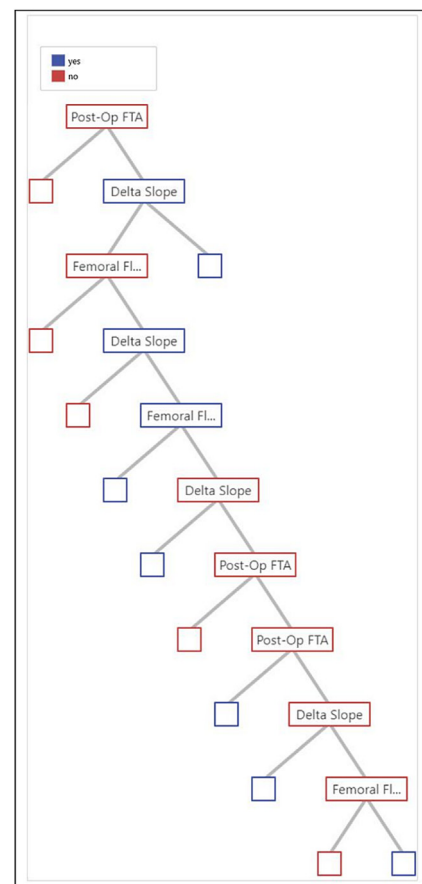


Fig. 3. Example of a classification and regression tree (CART) model showing the multiple interactive effect of coronal and sagittal alignment on achievement of MCIDs.

Table 1
Summary of Relative Importance (RI)-Scores for Classification and Regression Tree (CART) Models Based on Preoperative Tibio-Femoral Alignment (TFA) Groups.

Model Description	Preoperative Valgus	Preoperative Neutral	Preoperative Varus
MCID UCLA activity level (yes versus no)	*	ΔPTS = 100	ΔPTS = 100 FF = 55 TFA = 51
MCID KSS pain with level walking (yes versus no)	*	ΔPTS = 100 FF = 60	*
MCID KSS pain with climbing stairs (yes versus no)	FF = 100 TFA = 46 ΔPTS = 38	*	*
KSS 'knee normal score' (always versus not)	FF = 100 ΔPTS = 81 TFA = 8	ΔPTS = 100 TFA = 86 FF = 64	TFA = 100
MCID KOOS JR total (yes versus no)	TFA = 100	*	*
Satisfaction (very satisfied/satisfied versus not)	*	ΔPTS = 100 TFA = 88 FF = 65	*
MCIDs for KSS pain scores (yes versus no)	*	PTS = 100 FF = 60	*
MCIDs for KSS pain scores + KOOS JR (yes versus no)	TFA = 100	*	ΔPTS = 100 FF = 79
MCIDs for UCLA + KSS pain scores + KOOS JR (yes versus no)	TFA = 100	TFA = 100	*
MCIDs for UCLA + KSS pain scores + KOOS JR + satisfaction (yes versus no)	TFA = 100 ΔPTS = 46 FF = 27	TFA = 100 ΔPTS = 5 FF = 1	ΔPTS = 100 FF = 96 TFA = 4
MCIDs for UCLA + KSS pain scores + KOOS JR + satisfaction + KSS knee normal (yes versus no)	TFA = 100 ΔPTS = 60 FF = 47	TFA = 100 FF = 8	TFA = 100 FF 94 ΔPTS = 89

KSS pain scores, refers to both pain with level walking and pain with climbing stairs.

Asterisk (*) indicates no analysis could be performed (ie, the optimal tree was the root node).

TFA, postoperative tibio-femoral alignment unless designated as preoperative.

ΔPTS, change in posterior tibial slope from preoperative to postoperative.

FF, postoperative femoral component flexion.

MCID, minimal clinically important difference; KSS, Knee Society Score; UCLA, University of California Los Angeles; KOOS JR, Knee Injury and Osteoarthritis Outcome Score for Joint Replacement.

Results

All CART models are summarized in Table 1. Postoperative TFA and the change from preoperative to postoperative PTS were the most predictive for achieving multiple MCIDs in 90% of all CART models with mean RI scores of 89 and 71, respectively, followed by FF with a mean RI score of 52 (Table 2). A total of 57, 78, and 80% of models evaluating preoperative valgus, neutral, and varus-aligned knees had two or more interactions among postoperative TFA, change in PTS, and/or FF (Fig. 4). On average, preoperative valgus and neutral-aligned knee models had the best performance, with reasonably high areas under the receiver operating characteristic (ROC) curves for most models (mean 0.74 for both; range, 0.64–0.89) followed by preoperative varus knee models (mean 0.68; range, 0.55–0.78), which had slightly worse performance.

Regardless of the preoperative TFA group, approximating PTS within 4° of native PTS and slight to moderate FF (–2 to +7°) correlated with achieving multiple MCIDs and superior PROM scores (Table 3). However, optimal postoperative TFA differed by preoperative TFA group. Preoperative varus and neutral-aligned

knees were more likely to meet MCIDs and superior PROMs when not overcorrected into valgus postoperatively ($\geq -7^\circ$).

Preoperative valgus-aligned knees were more likely to achieve multiple MCIDs simultaneously when postoperative TFA was not overcorrected into substantial varus ($<0^\circ$). Of the preoperative valgus knees, 95% met MCIDs for KSS pain with level walking, KSS pain with climbing stairs, Knee Injury and Osteoarthritis Outcome Score for Joint Replacement, UCLA Activity Level, and reported optimal satisfaction and knees “always feeling normal” when postoperative TFA was not overcorrected to varus (Fig. 5). Specifically, area under the ROC learn and test curves for the preoperative valgus knee model achieving all MCIDs and optimal PROMs responses was 0.88 and 0.76, respectively (Fig. 6).

Table 2
Mean CART Model Relative Importance (RI) Scores.

Measurement	Mean RI Score
Postoperative TFA	89.3
Change in PTS	71.4
Postoperative FF	51.6

Higher RI-score = higher predictive ability on the outcome.

CART, classification and regression trees; TFA, tibiofemoral alignment; PTS, posterior tibial slope; FF, femoral component flexion.

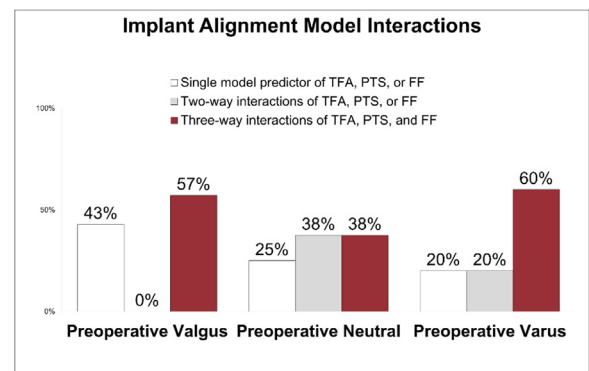


Fig. 4. Multiple interactions were observed among the 20 models evaluating achievement of MCIDs. Two- or three-way interactions were observed in at least 57% of all CART models highlighting the importance of evaluating coronal and sagittal alignment together.

Table 3

Summary of Optimal Alignment Zone Ranges Correlating With Individual and Simultaneous Achievement of MCIDs for All PROMs.

Preoperative TFA Alignment Group	Postoperative TFA	Change in PTS	Postoperative FF
Preoperative Valgus	−7 to 0	−4 to +1	+1 to +7
Preoperative Neutral	>−7	−4 to +4	<+3
Preoperative Varus	−7 to 0	−4 to −1	>−2

MCID, minimal clinically important difference; PROMs, patient-reported outcome measures; TFA, tibiofemoral alignment; PTS, posterior tibial slope; FF, femoral component flexion.

Discussion

The main findings of the current study were that approximating native tibial slope and similar TFA postoperatively with slight to moderate postoperative flexion of the femoral component correlated with achieving multiple MCIDs in modern PROMs. In addition, this study highlights the important interactive effect of coronal and sagittal alignment on achievement of PROM MCIDs, making a substantial advancement toward three-dimensional component positioning targets in TKA. To the authors' knowledge, this is the first study to evaluate the effect of both coronal and sagittal implant alignment together on PROMs. Current literature focuses on comparisons of mechanical and kinematic coronal alignment of knees [11,14–16]. More recent literature has identified non-arthritis and arthritic knee classification systems related to knee phenotypes [12,13], which has brought about the idea of a more personalized approach to alignment in TKA compared to the current systematic alignment algorithms of mechanical or kinematic alignment for all patients. The concept of knee phenotypes is intriguing, as the overwhelming majority of evidence comparing variations of mechanical and kinematic alignment strategies shows no difference in PROMs or survivorship out to 5 years [14,15]. In addition, neither technique has observed a substantial increase in satisfaction beyond the quoted 80%. However, to the author's knowledge, no studies to date have evaluated the effect of knee phenotypes on knee-specific PROMs, which may prove differently.

There are several potential limitations to the current study that should be acknowledged. The retrospective nature of this study may introduce inherent bias that a prospective randomized controlled trial may overcome. The sample size of the current study used for CART models was lower than ideal compared to sample sizes typically used for machine learning models. However, area under the ROC curve was good to great but not excellent, which highlights the complexity of alignment in multiple planes and correlation with PROMs. Data overfitting with machine learning models was apparent with some models and should be acknowledged. In addition, only a limited set of radiographic measurements

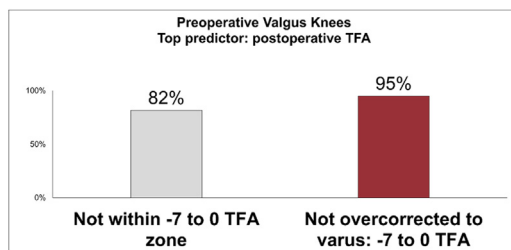


Fig. 5. Of the preoperative valgus knees, 95% achieved all MCIDs for Knee Society Score (KSS) pain scores, KOOS JR, UCLA activity level, and optimal responses for satisfaction and KSS knee feels 'always' normal when not overcorrected to substantial varus alignment postoperatively.

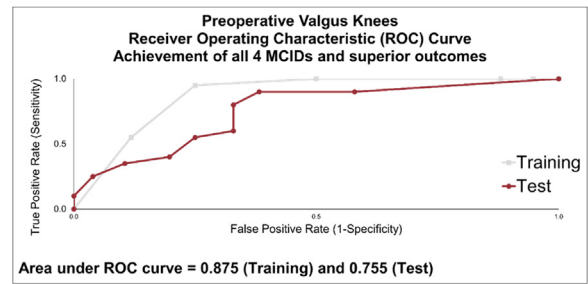


Fig. 6. Areas under the receiver operating characteristic (ROC) curves were similar for the training (learn curve) and test curve demonstrating that patient-reported outcome measures (PROMs) for preoperative valgus knees could be reasonably predicted from the CART model based on postoperative TFA, change in PTS, and postoperative FF component.

on short-leg radiographs performed by a single rater were used to evaluate coronal and sagittal alignment. However, measurement reliability was performed for a subset of cases to verify the accuracy of measurements. Other radiographic measurements (eg, femoral condylar offset) not included in this study may improve CART model performance and may affect the identified tibial slope alignment targets. Additionally, the results of this study are only pertinent to anterior-lipped and conforming bearing TKA designs, as no posterior stabilized designs were included in this study given the low sample size used by the treating surgeon. Furthermore, study results reflect the experience of a high-volume fellowship-trained arthroplasty surgeon at a single institution, which may limit the study's external validity. Future studies should replicate the methodology of this study to potentially validate these observed results at other institutions with more surgeons and surgical techniques.

Coronal alignment after TKA has been studied in depth with mixed results. Some studies demonstrate coronal alignment to be a major factor in satisfaction following TKA. In particular, postoperative malalignment of TKA components has been shown to lead to worse outcomes and early revision [26–28]. Ritter et al. suggested that knees should be aligned in neutral or slight amount of valgus to improve survivorship. Conversely, Meneghini et al. found that leaving varus knees in residual varus may not improve outcomes and pain [7]. Still, others report that patients in residual varus were observed to obtain higher KSS and Western Ontario and McMaster University Osteoarthritis Index scores compared to knees aligned neutrally or significantly greater varus at average follow-up of 7.2 years [29].

In addition, few studies provide data on the effect of sagittal component alignment on PROMs. One study by Farooq et al. [9] investigated the effect of sagittal alignment on optimal PROM scores in 1,091 TKAs. The authors found optimal sagittal implant position for maintaining native PTS and slight to moderate postoperative FF to correlate with greater satisfaction and more knees 'feeling always normal,' corroborating the current study findings. While alignment following TKA has been extensively studied, the majority of studies fail to control for preoperative alignment of the knees, which could lend to the mixed results observed in the literature. In fact, the native alignment of the knee varies widely among healthy non-arthritis and arthritic knees and by patient sex [12,30]. Jenny et al. recently reported that osteoarthritic (OA) knees had a wider distribution of coronal alignment compared to non-OA control knees. In addition, women tended to be aligned in more valgus compared to men in OA knees; however, there was still a wide distribution of women in both varus and valgus alignment. Given the recent published data on native knee and preoperative knee alignment, using a "one size fits all" systematic approach to all

TKAs may not lead to optimal patient outcomes [31]. The results of the current study corroborate these studies and suggest that improved outcomes may be achieved with certain coronal and sagittal implant alignments dependent on preoperative alignment.

Furthermore, despite persistent advancements in technologies such as robotics and computer navigation, approximately 20% of patients remain unsatisfied following TKA [32,33]. The use of these technologies has shown to reduce outliers related to implant alignment and therefore increase precision of implant placement. Song et al. [18] report that implant placement precision can be improved with robotic-assisted TKA. However, no study to date has defined true implant alignment targets for input into these advanced technologies, which calls into question the utility of these advanced technologies given the relatively steep learning curve and high cost [17]. Recently, Bush et al. [34] showed in a cohort of medial unicompartmental knee arthroplasties that surgeon experience may negate the need for robotic assistance by demonstrating an experienced surgeon using a manual technique had similar or higher precision of tibial component alignment in coronal and sagittal planes compared to published studies using robotic assistance. Nevertheless, robotic assistance could prove useful once optimal implant targets are rigorously studied and defined. Even with the tremendous developments in computer and robotic-assisted technology, no studies demonstrate supporting evidence of a comprehensive analysis identifying targets for coronal and sagittal alignment that can lead to improved outcomes. The current study provides insight into optimal TKA component alignment which evaluated both coronal and sagittal alignment of modern TKA implants. In addition, the current study used contemporary knee-specific PROMs as the primary outcome, which may prove beneficial for defining alignment targets in manual, computer navigation, and robotic-assisted TKA in the future.

In summary, optimized PROM MCIDs correlated with approximating native PTS while maintaining similar preoperative TFA and incorporating moderate FF. Study findings demonstrate interactions between sagittal and coronal alignment parameters to optimize PROMs, highlighting the importance of multiplanar implant alignment targets. In addition, this is the first study, to the authors' knowledge, to report optimal TKA component position in coronal and sagittal planes correlating with MCIDs in PROMs and the first study to report on postoperative TKA alignment goals for preoperative valgus knees. While further study with larger numbers is warranted, our results have the potential to support alignment targets and optimize PROMs following TKA using both manual and computer-assisted techniques.

References

- [1] Dunbar MJ, Richardson G, Robertsson O. I can't get no satisfaction after my total knee replacement: rhymes and reasons. *Bone Joint J* 2013;95-B:148–52. <https://doi.org/10.1302/0301-620X.95B11.32767>.
- [2] Kahlenberg CA, Nwachukwu BU, McLawhorn AS, Cross MB, Cornell CN, Padgett DE. Patient satisfaction after total knee replacement: a systematic review. *HSS J* 2018;14:192–201. <https://doi.org/10.1007/s11420-018-9614-8>.
- [3] Nakano N, Shoman H, Olavarria F, Matsumoto T, Kuroda R, Khanduja V. Why are patients dissatisfied following a total knee replacement? A systematic review. *Int Orthop* 2020;44:1971–2007. <https://doi.org/10.1007/s00264-020-04607-9>.
- [4] Farooq H, Deckard ER, Ziemba-Davis M, Madsen A, Meneghini RM. Predictors of patient satisfaction following primary total knee arthroplasty: results from a traditional statistical model and a machine learning algorithm. *J Arthroplasty* 2020;35:3123–30. <https://doi.org/10.1016/j.arth.2020.05.077>.
- [5] Khatib Y, Badge H, Xuan W, Naylor JM, Harris IA. Patient satisfaction and perception of success after total knee arthroplasty are more strongly associated with patient factors and complications than surgical or anaesthetic factors. *Knee Surg Sports Traumatol Arthrosc* 2020;28:3156–63. <https://doi.org/10.1007/s00167-019-05804-9>.
- [6] Gromov K, Korchi M, Thomsen MG, Husted H, Troelsen A. What is the optimal alignment of the tibial and femoral components in knee arthroplasty? *Acta Orthop* 2014;85:480–7. <https://doi.org/10.3109/17453674.2014.940573>.
- [7] Meneghini RM, Grant TW, Ishmael MK, Ziemba-Davis M. Leaving residual varus alignment after total knee arthroplasty does not improve patient outcomes. *J Arthroplasty* 2017;32:S171–6. <https://doi.org/10.1016/j.arth.2017.02.064>.
- [8] Peng Y, Arauz P, An S, Limmahakhun S, Klemm C, Kwon Y-M. Does component alignment affect patient reported outcomes following bicruciate retaining total knee arthroplasty? An in vivo three-dimensional analysis. *J Knee Surg* 2020;33:798–803. <https://doi.org/10.1055/s-0039-1688500>.
- [9] Farooq H, Deckard ER, Arnold NR, Meneghini RM. Machine learning algorithms identify optimal sagittal component position in total knee arthroplasty. *J Arthroplasty* 2021;36:S242–9. <https://doi.org/10.1016/j.arth.2021.02.063>.
- [10] Rivière C, Iranpour F, Auvinet E, Howell S, Vendittoli P-A, Cobb J, et al. Alignment options for total knee arthroplasty: a systematic review. *Orthop Traumatol Surg Res* 2017;103:1047–56. <https://doi.org/10.1016/j.otsr.2017.07.010>.
- [11] Courtney PM, Lee G-C. Early outcomes of kinematic alignment in primary total knee arthroplasty: a meta-analysis of the literature. *J Arthroplasty* 2017;32:2028–2032.e1. <https://doi.org/10.1016/j.arth.2017.02.041>.
- [12] Hirschmann MT, Moser LB, Amsler F, Behrend H, Leclercq V, Hess S. Functional knee phenotypes: a novel classification for phenotyping the coronal lower limb alignment based on the native alignment in young non-osteoarthritic patients. *Knee Surg Sports Traumatol Arthrosc* 2019;27:1394–402. <https://doi.org/10.1007/s00167-019-05509-z>.
- [13] MacDessi SJ, Griffiths-Jones W, Harris IA, Bellemans J, Chen DB. Coronal plane alignment of the knee (CPAK) classification. *Bone Joint J* 2021;103-B:329–37. <https://doi.org/10.1302/0301-620X.103B2.BJJ-2020-1050.R1>.
- [14] Young SW, Walker ML, Bayan A, Briant-Evans T, Pavlou P, Farrington B. The chitranjan S. Ranawat award: No difference in 2-year functional outcomes using kinematic versus mechanical alignment in TKA: a randomized controlled clinical trial. *Clin Orthop Relat Res* 2017;475:9–20. <https://doi.org/10.1007/s11999-016-4844-x>.
- [15] Young SW, Sullivan NPT, Walker ML, Holland S, Bayan A, Farrington B. No difference in 5-year clinical or radiographic outcomes between kinematic and mechanical alignment in TKA: a randomized controlled trial. *Clin Orthop Relat Res* 2020;478:1271–9. <https://doi.org/10.1097/CORR.0000000000001150>.
- [16] Sappey-Marinière E, Pauvert A, Batailler C, Swan J, Cheze L, Servien E, et al. Kinematic versus mechanical alignment for primary total knee arthroplasty with minimum 2 years follow-up: a systematic review. *SICOT J* 2020;6:18. <https://doi.org/10.1051/sicotj/2020014>.
- [17] Kayani B, Konan S, Ayuob A, Onochie E, Al-Jabir T, Haddad FS. Robotic technology in total knee arthroplasty: a systematic review. *EFORT Open Rev* 2019;4:611–7. <https://doi.org/10.1302/2058-5241.4.190022>.
- [18] Song E-K, Seon J-K, Yim J-H, Netravali NA, Bargar WL. Robotic-assisted TKA reduces postoperative alignment outliers and improves gap balance compared to conventional TKA. *Clin Orthop Relat Res* 2013;471:118–26. <https://doi.org/10.1007/s11999-012-2407-3>.
- [19] Meneghini RM, Mont MA, Backstein DB, Bourne RB, Dennis DA, Scuderi GR. Development of a modern knee society radiographic evaluation system and methodology for total knee arthroplasty. *J Arthroplasty* 2015;30:2311–4. <https://doi.org/10.1016/j.arth.2015.05.049>.
- [20] Copay AG, Eyberg B, Chung AS, Zurcher KS, Chutkan N, Spangehl MJ. Minimum clinically important difference: current trends in the orthopaedic literature, Part II: lower extremity: a systematic review. *JBJS Rev* 2018;6:e2. <https://doi.org/10.2106/JBJS.RVW.17.00160>.
- [21] Lee WC, Kwan YH, Chong HC, Yeo SJ. The minimal clinically important difference for Knee Society Clinical Rating System after total knee arthroplasty for primary osteoarthritis. *Knee Surg Sports Traumatol Arthrosc* 2017;25:3354–9. <https://doi.org/10.1007/s00167-016-4208-9>.
- [22] Lyman S, Lee Y-Y, McLawhorn AS, Islam W, MacLean CH. What are the minimal and substantial improvements in the HOOS and KOOS and JR versions after total joint replacement? *Clin Orthop Relat Res* 2018;476:2432–41. <https://doi.org/10.1097/CORR.0000000000000456>.
- [23] Friedman JH, Meulman JJ. Multiple additive regression trees with application in epidemiology. *Stat Med* 2003;22:1365–81. <https://doi.org/10.1002/sim.1501>.
- [24] Breiman L. Classification and Regression Trees. New York: Routledge; 2017. <https://doi.org/10.1016/9781315139470>.
- [25] Friedman JH. Greedy function approximation: a gradient boosting machine. *Ann Stat* 2001;29:1189–232.
- [26] Ritter MA, Faris PM, Keating EM, Meding JB. Postoperative alignment of total knee replacement. Its effect on survival. *Clin Orthop Relat Res* 1994;299:153–6.
- [27] Hadi M, Barlow T, Ahmed I, Dunbar M, McCulloch P, Griffin D. Does malalignment affect patient reported outcomes following total knee arthroplasty: a systematic review of the literature. *Springer Plus* 2016;5:1201. <https://doi.org/10.1186/s40064-016-2790-4>.
- [28] Kazarian GS, Haddad FS, Donaldson MJ, Wignadasan W, Nunley RM, Barrack RL. Implant malalignment may be a risk factor for poor patient-reported outcomes measures (PROMs) following total knee arthroplasty (TKA). *J Arthroplasty* 2022;37:S129–33. <https://doi.org/10.1016/j.arth.2022.02.087>.
- [29] Vanlommel L, Vanlommel J, Claes S, Bellemans J. Slight undercorrection following total knee arthroplasty results in superior clinical outcomes in varus knees. *Knee Surg Sports Traumatol Arthrosc* 2013;21:2325–30. <https://doi.org/10.1007/s00167-013-2481-4>.
- [30] Jenny J-Y, Baldairon F, Hirschmann MT. Functional knee phenotypes of OA patients undergoing total knee arthroplasty are significantly more varus

- or valgus than in a non-OA control group. *Knee Surg Sports Traumatol Arthrosc* 2022;30:2609–16. <https://doi.org/10.1007/s00167-021-06687-5>.
- [31] Hirschmann MT, Behrend H. Functional knee phenotypes: a call for a more personalised and individualised approach to total knee arthroplasty? *Knee Surg Sports Traumatol Arthrosc* 2018;26:2873–4. <https://doi.org/10.1007/s00167-018-4973-8>.
- [32] Gunaratne R, Pratt DN, Banda J, Fick DP, Khan RJK, Robertson BW. Patient dissatisfaction following total knee arthroplasty: a systematic review of the literature. *J Arthroplasty* 2017;32:3854–60. <https://doi.org/10.1016/j.arth.2017.07.021>.
- [33] Batailler C, Swan J, Sappey Marinier E, Servien E, Lustig S. New technologies in knee arthroplasty: current concepts. *J Clin Med* 2020;10:E47. <https://doi.org/10.3390/jcm10010047>.
- [34] Bush AN, Ziemba-Davis M, Deckard ER, Meneghini RM. An experienced surgeon can meet or exceed robotic accuracy in manual unicompartmental knee arthroplasty. *J Bone Joint Surg Am* 2019;101:1479–84. <https://doi.org/10.2106/JBJS.18.00906>.